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[ SYSTEM ] :: initialising heterogeneous ensemble · 4 runs · 99,999 → 499,995 flows · seed=42
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A Comparative Evaluation of Heterogeneous Ensemble Voting & Stacking Strategies for Multi-Class Flow-Based DDoS Detection

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RESEARCH QUESTION

"Which heterogeneous ensemble strategy is most effective for flow-based multi-class DDoS detection?"

Hypothesis: No single strategy dominates across all metrics – **stability over victory** determines which is appropriate for production.

9 classes

8 DDoS VECTORS + BENIGN + HPING3 + IPERF

499,995

FLOWS TESTED AT PRODUCTION SCALE

4 runs

A · B · C · D – FEATURE × SCALE MATRIX

01 / AIMS & CONTEXT

What is a DDoS flood?

THE THREAT LANDSCAPE

Cloudflare mitigated 47.1M DDoS attacks in 2025 – double the year before, averaging 5,376 per hour. Attackers coordinate botnets via a single command host to flood victims across TCP, UDP & ICMP.

Signature-based defences cannot keep up with non-linear, multi-protocol traffic. Machine learning classifies flows directly – but no single model captures every pattern, which is why ensembles exist.

PROJECT AIMS & OBJECTIVES

- Construct a balanced multi-class flow dataset of 8 DDoS attacks + benign traffic
- Benchmark 4 diverse base learners under identical conditions
- Compare Voting vs Stacking strategies across 4 feature × scale runs
- Identify the strategy most stable for production deployment

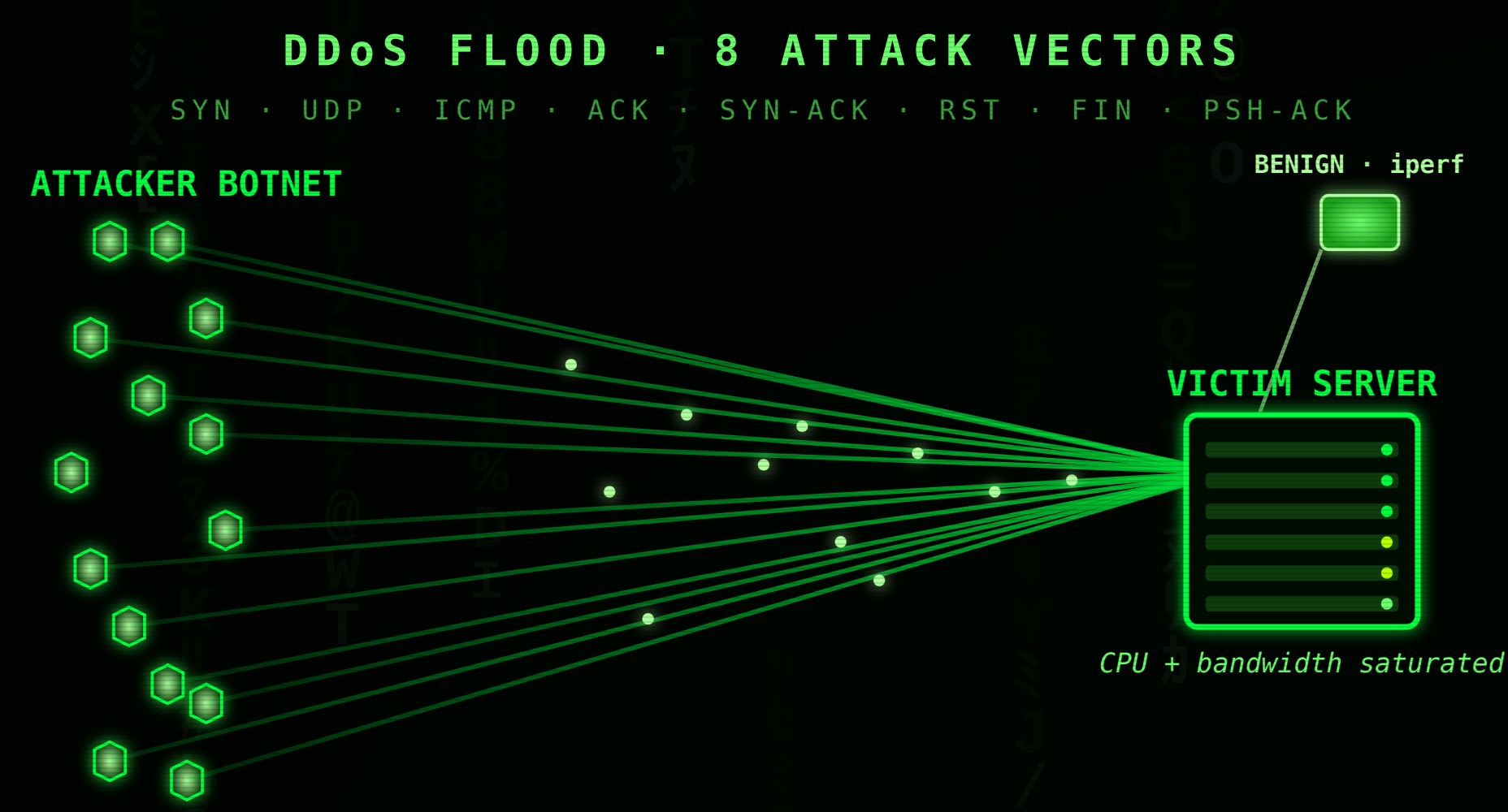


FIG 1 · 8 DDoS vectors flood victim server. Benign iperf traffic competes for bandwidth.

02 / METHODOLOGY · DATA

Flow-based pipeline

MULTI-CLASS · BALANCED · REPRODUCIBLE

Three Ubuntu VMs in an isolated 10.0.0.0 network. Two hping3 attackers fire SYN, UDP, ICMP, ACK, SYN-ACK, RST, FIN & PSH-ACK at a victim, while iperf generates benign baseline traffic.

A custom Scapy extractor converts pcaps into **bidirectional flow records** with 27 features (packet counts, byte counts, IATs, rates, TCP flags). IPs removed to prevent memorisation. Classes **undersampled to 11,111 each** for a balanced 99,999-flow dataset.

TOOLS Ubuntu 22.04 hping3 iperf tcpdump Scapy Python 3.11

CONSTRUCTION PIPELINE

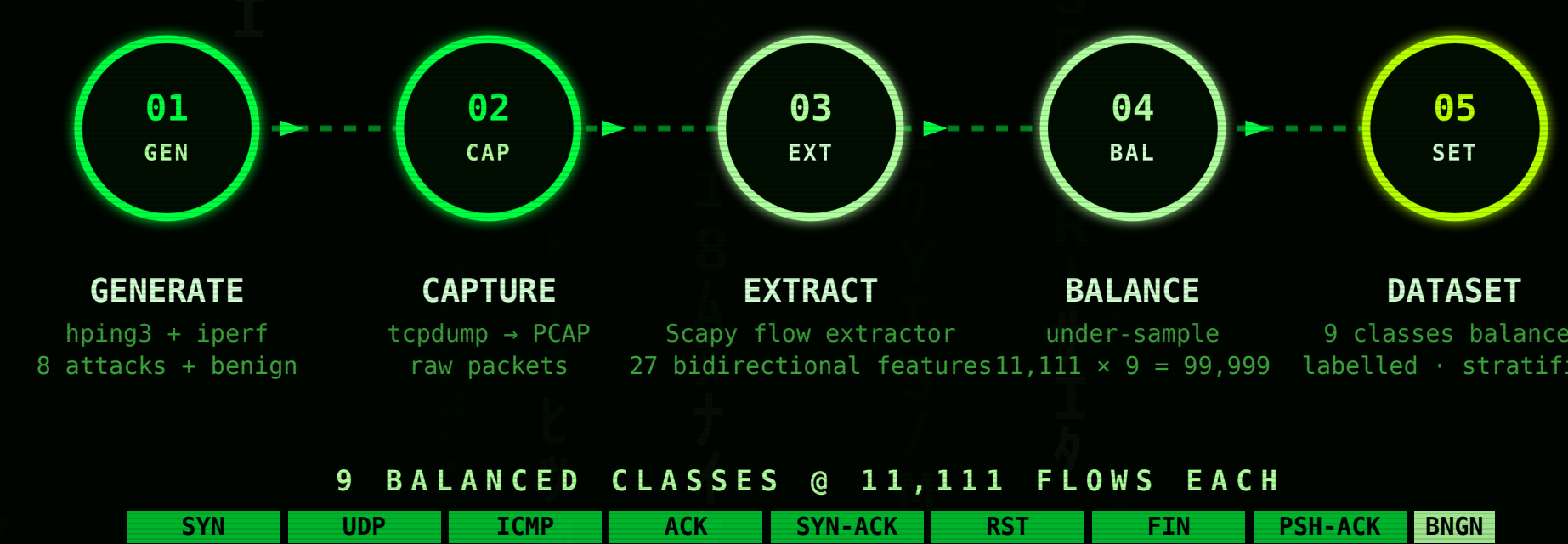


FIG 2 · 5-stage pipeline: traffic generation → capture → flow extraction → balancing → labelled dataset.

03 / METHODOLOGY · MODEL

Heterogeneous ensemble

4 BASE LEARNERS · 4 STRATEGIES · HEAD-TO-HEAD

Eight baselines ranked by a **weighted performance score** (F1-Macro > MCC > Recall > Precision > Accuracy). Top four picked for **learning diversity**:

- Decision Tree – rule-based non-linear
- Logistic Regression – linear baseline
- SVM – maximum-margin separation
- MLP – neural non-linear patterns

TOOLS scikit-learn pandas NumPy Azure ML 5-fold CV McNemar χ^2

ENSEMBLE ARCHITECTURE

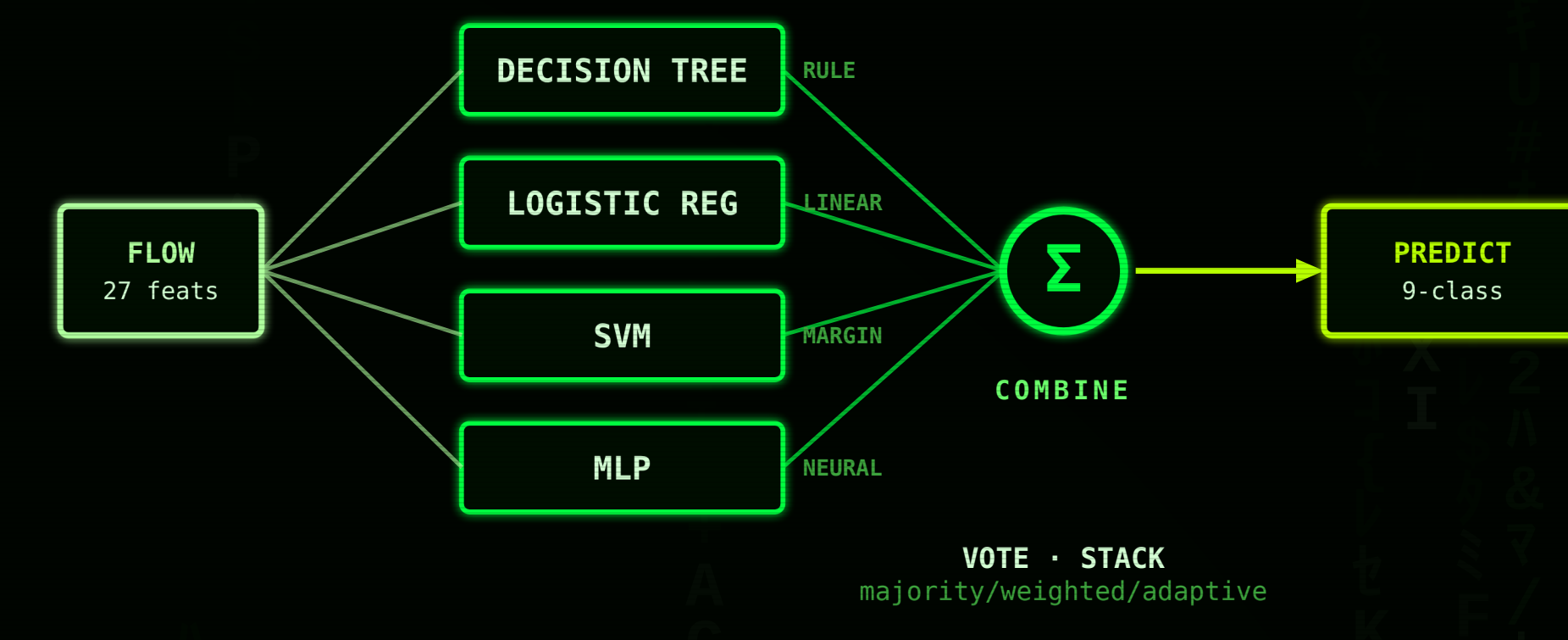


FIG 3 · Heterogeneous ensemble: 4 diverse base learners combined via 4 strategies.

04 / KEY LITERATURE Summary of foundational work

WHY HETEROGENEOUS ENSEMBLES & WHY F1-MACRO

Dietterich (2009)
Established **ensembles outperform single learners** when base models are accurate AND diverse – the foundational case for heterogeneous over homogeneous.

Hasanah et al. (2024)
Compared voting strategies on flow-based intrusion detection but used **imbalanced data** and accuracy – gap this project fills with balanced multi-class data.

Verma et al. (2024)
Demonstrated **stacking's meta-learning advantage** on cyber traffic, but evaluated only one stacking config and no head-to-head with voting.

Angulo et al. (2026)
Most recent multi-class DDoS benchmark – confirms **F1-Macro is the fair metric** for class-balanced multi-class flow detection.

Hancock et al. (2023)
Surveyed evaluation pitfalls in network IDS – found **accuracy misleads** when classes are even mildly imbalanced or when error costs differ.

Grandini et al. (2020)
Defined the **multi-class metric hierarchy** used here: F1-Macro > MCC > Recall > Precision > Accuracy for balanced multi-class problems.

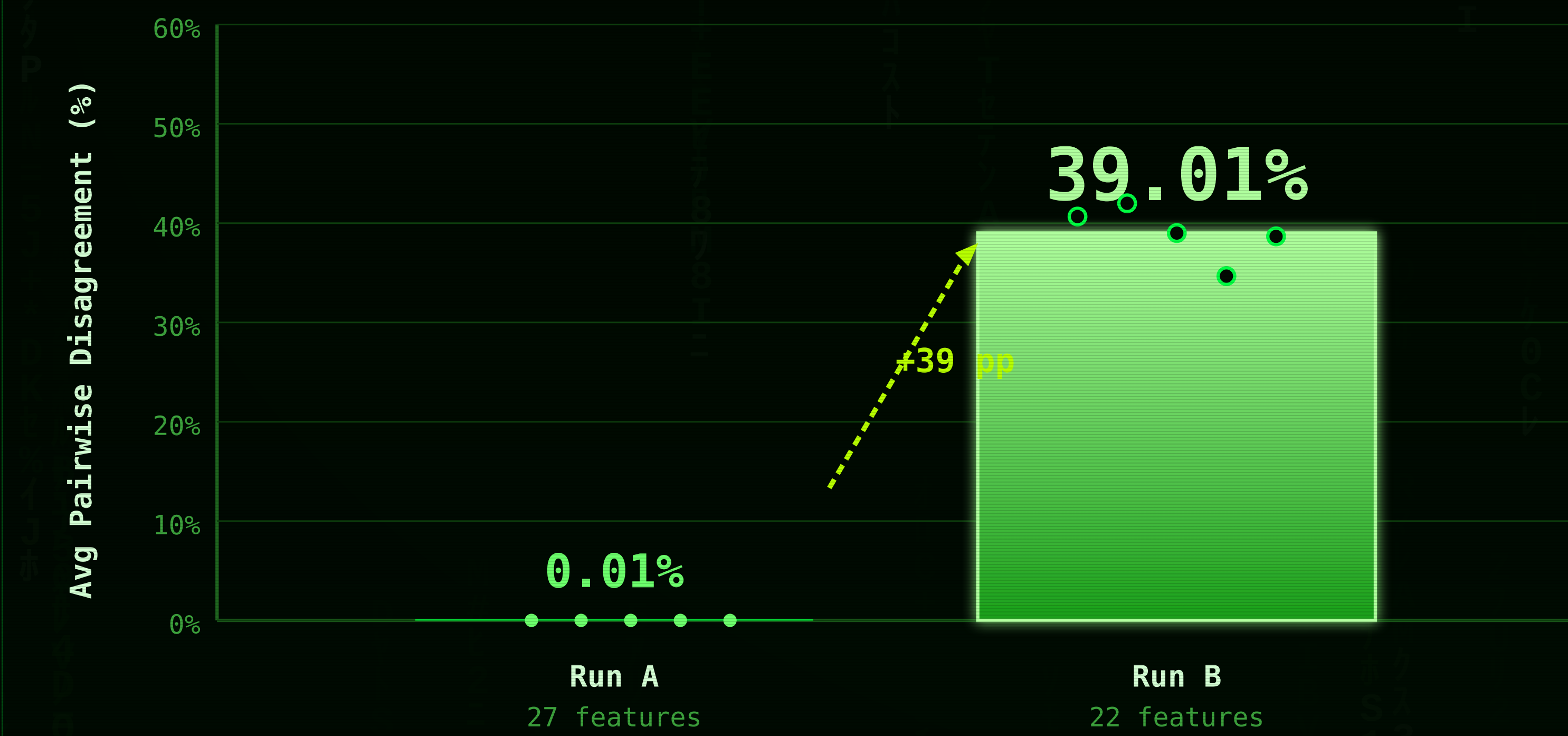
Pandey et al. (2024)
Used **5-fold stratified CV** for flow-based DDoS – methodology this project adopts and extends to 4 feature × scale runs.

Cloudflare (2026)
Reports **47.1M DDoS attacks mitigated in 2025**, doubled YoY – establishes urgency of robust multi-protocol detection.

05 Results · the heterogeneity collapse

FEATURE REDUCTION DRIVES DISAGREEMENT · NOT SCALE

PAIRWISE DISAGREEMENT – RUN A vs RUN B
4 BASE MODELS · 99,999 FLOWS · 5-FOLD STRATIFIED CV



> TCP FLAG REMOVAL UNLEASHES HETEROGENEITY

Disagreement explodes from 0.01% → 39.01% when TCP flag counts are removed. Heterogeneity was *hidden*, not absent.

> RUN A · 27 features
All four strategies **tie at F1-Macro = 0.9634**. Disagreement collapses to 0.0%. Clean feature set hides heterogeneity.

> RUN B · 22 features
Strip TCP flags, ensembles split. All degrade by Δ F1-Macro ≈ -0.55 – Stacking absorbs new disagreement best.

> RUN C · 499,995 flows
Scale alone doesn't separate strategies. But Stacking's **fold-variance is 24% lower** than any voter.

> RUN D · scaled + reduced
Disagreement holds at 37.78% – within 1.23% of Run B at 5× the data. **Features drive heterogeneity.**

> **McNemar's test confirms the collapse is structural.** All four strategies show $\chi^2 > 52,000$ ($p < 0.001$) between Run A and Run B. The 22-feature set produced zero unique correct predictions that Run A didn't already cover – the five removed TCP flag features carried decisive information that **no ensemble strategy could compensate for**.

CROSS-RUN STRATEGY DELTA (F1-MACRO)

> Δ F1-MACRO FROM RUN A → RUN B · BRIGHTEST = MOST ROBUST

STRATEGY	RUN A · F1-MACRO	RUN B · F1-MACRO	Δ F1-MACRO
Majority Vote	0.9634	0.4131	-0.5503
Adaptive Vote	0.9634	0.4172	-0.5462
Weighted Vote	0.9634	0.4171	-0.5463
STACKING	0.9634	0.4219	-0.5415

> STACKING DEGRADES LEAST WHEN FEATURES ARE STRIPPED

Stacking has the smallest $|\Delta$ F1-Macro| (-0.5415) vs Majority Vote's -0.5503. At scale, std = 0.000235 vs voters 0.000307 = **24% less variance**.

06 Discussion · what this means in production

THREE DEPLOYMENT-LEVEL RECOMMENDATIONS

DESIGN

Engineer features before scaling data

Run C scaled flows 5× but rankings held. Run B's feature reduction caused the real movement. **Diversity lives in the feature space**, not the row count – invest engineering effort there first.

EVALUATION

Weight metrics away from accuracy

On Run B, accuracy looks recoverable at 43.86%, but F1-Macro = 0.4131 and MCC = 0.3848 show the true drop. A **weighted score** (F1-Macro > MCC > Recall > Precision > Accuracy) ranks fairly on imbalanced multi-class flows.

DEPLOYMENT

Stacking for stability, Voting for cost

Pick **Stacking** when reliability across shifting traffic matters most – meta-learning absorbs disagreement and varies 24% less per fold. Pick **Voting** when compute budget is the binding constraint.

07 / CONCLUSION · KEY FINDING

Choose stability over victory.

No single ensemble dominates on F1-Macro alone – all four **tie at 4 d.p.** on the clean 27-feature dataset. When feature reduction forces real disagreement, **Stacking** handles it best through meta-learning, and at production scale it varies **24% less** across folds than any voter.

For deployable DDoS detection where reliability matters more than a fractional F1-Macro edge, **Stacking is the recommended strategy**. Voting remains the lower-cost alternative when compute is constrained.

- Validate on CIC-DDoS2019 & zero-day attacks
- Test Adaptive Voting on live streaming traffic
- Explore hybrid ML / DL ensemble compositions

GAP 1 Head-to-head ensemble comparison under identical conditions

GAP 2 Stability assessment across 4 feature × scale runs

GAP 3 Per-use-case strategy guidance for production IDS

> **STABILITY OVER VICTORY** · Built on Azure ML · 5-fold stratified CV · seed = 42 · github.com/dan-kinsiona/ddos-ensemble

Cloudflare 2026 Threat Report · Dietterich (2009) · Hasanah et al. (2024) · Verma et al. (2024) · Angulo et al. (2026) · Hancock et al. (2023) · Grandini et al. (2020) · Pandey et al. (2024)

> **REFERENCE LIST**